CS315 · ALGORITHMS ANALYSIS & DESIGN

Practice 3 From the 461 Final

Solutions

Only the 461 final questions that fall within the midterm

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01 Solution Q1: MCQ

#	Answer	Reason
3	**c. Collision**	The definition of a collision: different elements mapped to the same cell.
4	**b**	Chaining does not avoid a collision. It handles one after it happens. The other three reduce the chance of it happening.

02 Solution Q2: short answers

A) Divide, then conquer, then combine: break the problem into sub-problems of the same form, solve them recursively, and combine the solutions.

B) Merge sort and quick sort.

Asked	Rule for order m	m = 7
Max number of child	m	**7**
Max number of key	$m - 1$	**6**
Min number of key	$\lceil \frac{m}{2} \rceil - 1$	**3**

The minimum number of keys is 3 because the minimum number of children is $\lceil \frac{7}{2} \rceil = 4$, and a node always has one key fewer than children. This holds for every node other than the root. The root needs only one key.

03 Solution Q3: analysis

Line	# operations	Why
`1. w ← 0`	1	An assignment.
`2. c ← 1`	1	An assignment.
`3. while w < n² do`	$n^2 + 1$	The loop runs n^2 times, and the test runs once more when it fails.
`4. w ← w + 1`	$2n^2$	An addition and an assignment.
`5. c ← c * a`	$2n^2$	A multiplication and an assignment.
`6. return c`	1	The return.
Total	$\mathbf{5n^2 + 4}$	$O(n^2)$

$$T(n) = 1 + 1 + (n^2 + 1) + 2n^2 + 2n^2 + 1 = 5n^2 + 4 \quad \rightarrow \quad O(n^2)$$

The algorithm multiplies `c` by `a` a total of n^2 times, so it returns `a` to the power n^2. It has no separate best and worst cases, because the number of iterations depends on `n` only.

If you count computing n^2 in every test as an extra operation, the total becomes $6n^2 + 5$. The result is the same: $O(n^2)$.

04 Solution Q4: B-tree

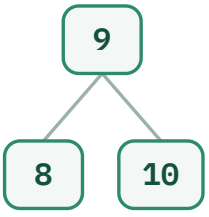
A B-tree of degree 3: at most 2 keys per node. When a node reaches 3 keys it splits and the middle key moves up. This is the definition confirmed by the grading of the 462 midterm.

1 Insert 8, 9

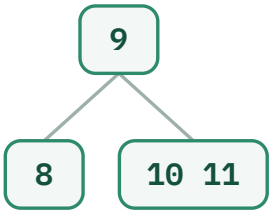


2 Insert 10

It becomes `8 9 10`: three keys, so it splits and 9 moves up.

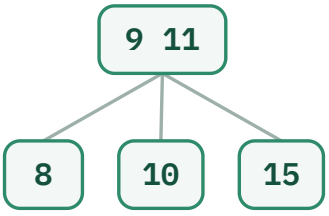


3 Insert 11



4 Insert 15

The leaf becomes `10 11 15`, so it splits and 11 moves up beside 9.



5 Insert 20 · the final tree

